

Linear algebra – 7

Euclidean spaces, orthogonality and the SVD

Homework

12 сентября 2026 года

Theoretical minimum. Self-adjoint, orthogonal and normal operators; the spectral theorem; singular values and the SVD; the minimax description of σ_k ; polar decomposition; the Moore–Penrose pseudoinverse; orthogonal projections and least squares.

1. Prove the existence of the singular value decomposition and describe the image of the unit ball; show that $\sigma_1 \geq |\lambda| \geq \sigma_n$ for every eigenvalue λ of a square matrix A .
2. (Eckart–Young 1936) Prove that among matrices of rank at most k the closest one to A both in the operator and in the Frobenius norm is obtained by keeping the k largest singular values, and that the distance equals σ_{k+1} (respectively $(\sum_{i>k} \sigma_i^2)^{1/2}$). Deduce that the distance from A to the set of singular matrices is σ_n .
3. (IMC) Let H be an infinite-dimensional real Hilbert space, $d > 0$, and let $S \subseteq H$ be a set (not necessarily countable) any two distinct points of which are at distance d . Show that there is $y \in H$ such that $\{\frac{\sqrt{2}}{d}(x - y) : x \in S\}$ is an orthonormal system.
4. (SEEMOUS 2023) Prove that if $A \in M_n(\mathbb{C})$ satisfies $A + A^* = A^2 A^*$, then $A = A^*$.
5. (IMC) A complex $n \times n$ matrix A is called t -normal if $AA^T = A^T A$. For each n determine the maximal dimension of a linear space of complex $n \times n$ matrices consisting of t -normal matrices.
6. (IMC) Does there exist a real 3×3 matrix A with $\operatorname{tr} A = 0$ and $A^2 + A^T = E$?

7. (SEEMOUS 2007) Let $f(x) = \max_{1 \leq i \leq n} |x_i|$ and let A be an $n \times n$ matrix with $f(Ax) = f(x)$ for all $x \in \mathbb{R}^n$. Prove that $A^m = E$ for some positive integer m .
8. (Mexmat, 2026) Let $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an orthogonal transformation with positive determinant. The *period* of a vector v is the least positive integer m with $A^m v = v$. Suppose there are s vectors with pairwise coprime periods greater than 1. What is the least possible n ?
9. Prove the polar decomposition and the uniqueness of $\sqrt{A^T A}$; then define the Moore–Penrose pseudoinverse through the SVD, check the four Penrose identities, and show that $A^+ b$ is the least-squares solution of smallest norm.
10. (Romania national olympiad 2024) Let $A, B \in M_n(\mathbb{R})$ and consider $f: M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$, $f(Z) = AZ + B\overline{Z}$, where $\overline{Z}$ is the entry-wise complex conjugate. Prove that the following are equivalent: (1) f is injective; (2) f is surjective; (3) both $A + B$ and $A - B$ are invertible.